HARNESSEVAL

Danh gia ha tang Coding Agent thong qua phan tich ablation da chieu tren nhieu LLM

Evaluating Coding Agent Scaffolds Through Ablation-Driven Multi-Dimensional Analysis

Phuong phap hoc tap & Nghien cuu khoa hoc

FPT School of Business and Technology (FSB)

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v2 Phien ban sau phan bien hoi dong

NHOM NGHIEN CUU

GIANG VIEN HUONG DAN

[Ten thay/co — xin cung cap]

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MUC LUC

1. **Dat van de** — nghich ly evaluation
2. **Research gap** — ba ly do
3. **Muc dich** & 5 muc tieu
4. **Gia thuyet** H1–H4
5. **Taxonomy** 3 chieu, 7 metrics
6. **Kien truc Modular Harness**
7. **Phuong phap** — 4 giai doan
8. **Ablation design** 27x150
9. **ANOVA 2 chieu** — phan tach harness vs LLM
10. **Threats & Risks**
11. **Timeline** & chi phi
12. **Trang thai code** — 167 tests, 9 phases
13. **Ket qua so bo** (ANOVA thuc nghiem)
14. **Dong gop** & publishability
15. **Q&A**

1. DAT VAN DE: NGHICH LY EVALUATION

Hieu suat **coding agent** phu thuoc 2 yeu to:

A. LLM co so

Claude, GPT, DeepSeek...

Da co nhieu benchmark danh gia

B. Harness / Scaffold

Tool system, context manager, orchestration...

"He dieu hanh cho agent" [A1]

Bang chung: Lou et al. (2026) [A2] cho thay model nho + harness tot co the vuot model lon + harness thuong, giam 100% hanh dong bat hop le. Bui et al. (2026) [A1]: Adaptive Context Compaction giam **54% token**.

Nghich ly: Khi bao cao "Agent A dat 70%," chung ta khong biet component nao cua harness dong gop vao ket qua do.

2. RESEARCH GAP: BA LY DO TON TAI THAT SU

Phan tich **49 papers** (2023–2026): khong co cong trinh nao danh gia ban than harness mot cach he thong [S5].

1. Vai tro khac nhau

Princeton (SWE-Agent), Anthropic (Claude Code) la developers — ho xay harness, khong phai evaluators so sanh he thong.

Analogy: hang xe vs Euro NCAP

2. Linh vuc chay dua

2023–2026: tu vai harness len hang chuc. Chua ai dung lai de meta-evaluate vi con dang xay dung.

3. Bang chung gian tiep

Lou et al. so sanh harness tren game tasks [A2]. Robeyns et al. tu cai thien harness nhung khong do tung component [B10].

Research Gap: Thieu khung danh gia cho phep (a) do chat luong tung thanh phan harness, (b) phan tach dong gop harness vs LLM, (c) so sanh cac harness tren chieu chat luong thong nhat.

2.1. CU THE: DA CO GI — CHUA CO GI

Khia canh	Da co	Chua co
Do output cua agent	SWE-bench, SWE-Compass, AgentBoard	—
Do tool accuracy rieng le	ToolLLM, AnyTool, Gorilla	Do tool TRONG harness coding
Do memory rieng le	Mem0, A-MEM (tren chatbot)	Do memory TRONG harness coding
Mo ta harness design	OpenDev, AutoHarness	Do luong harness quality
So sanh cross-backend	—	**Hoan toan chua ai lam**
Tach harness vs LLM	—	**Hoan toan chua ai lam**

3. MUC DICH + 5 MUC TIEU

Muc dich: Xay dung **khung danh gia tong hop dau tien** cho ha tang (harness) cua coding agent, cho phep phan tach va dinh luong dong gop tung thanh phan harness vao hieu suat agent, **doc lap voi LLM co so**.

MT	Noi dung	Tieu chi
MT1	Taxonomy 3 chieu, 7 metrics, validate boi 5 chuyen gia	Fleiss kappa ≥ 0.6
MT2	Fork & refactor SWE-Agent thanh modular harness	+/- 3% vs SWE-Agent goc
MT3	Ablation 150 tasks x 27 conditions (3x3x3), co pilot study	7,050 evaluations
MT4	ANOVA 2 chieu phan tach variance harness vs LLM	$p < 0.05$ Bonferroni
MT5	Phat hanh open-source toolkit tren GitHub	Docs + CI/CD

v2 Giam tu 5 chieu → **3 chieu**, 80 → **27 conditions**, them pilot study, them ANOVA 2 chieu (cau hoi cua hoi dong).

4. GIA THUYET NGHIEN CUU

H1

Tool system co effect size **lon nhat** (Cohen's d > 0.5) trong 3 thanh phan harness doi voi resolve rate.

H2

Context management co effect size **trung binh** (Cohen's d 0.3 – 0.5).

H3

Harness configuration giai thich **it nhat 20%** variance cua resolve rate ($eta^2 \geq 0.20$), doc lap voi LLM backend.

H4

Variance giua cac LLM backends **giam** khi harness chat luong cao hon (**interaction effect** co y nghia thong ke).

Trung tinh: Khong du doan con so ("30–40%"). Chi neu huong co the kiem chung → tranh confirmation bias. **Du H duoc ho tro hay bac bo deu la dong gop.**

5. TAXONOMY 3 CHIEU — 7 METRICS

Chieu	Metric	Dinh nghia	Cach do
D1 Tool Dispatch	**M1.1** Correct Selection Rate	% tool calls chon tool phu hop (tu **tap acceptable**)	2 annotators, Fleiss kappa
	M1.2 Redundant Call Rate	% calls ma output khong dung trong 3 turns sau	Log analysis tu dong
	M1.3 Utilization Breadth	unique tools used / total available	Log analysis tu dong
D2 Context Utilization	**M2.1** Info Retention	BERTScore giua full-context vs compacted output	30 cap human preference
	M2.2 Effective Token Ratio	% tokens lien quan task hien tai	LLM classifier (Haiku) + 200 human labels
D3 Backend Portability	**M3.1** Cross-Backend StdDev	std(RR) qua 3 LLM backends	Truc tiep
	M3.2 Min/Max Ratio	min(RR) / max(RR)	Gia tri cao = portable

v2 M1.1: doi "ground-truth duy nhat" → **tap acceptable tools** (cau hoi #4). M2.2: doi "token waste" mo ho → **LLM classifier + human validation** (cau hoi #9).

6. KIEN TRUC MODULAR HARNESS — 3 SWAPPABLE LAYERS

LAYER 1: Tool Dispatch — `ToolProvider`

Full (12 tools) | Medium (8) | Minimal (5)

LAYER 2: Context Management — `ContextStrategy`

Full History | Sliding Window (50K) | Summary/ACC (2K)

LAYER 3: LLM Backend — `LLMBackend`

Claude Sonnet 4 | GPT-4o | DeepSeek-V3

L4 Safety & Permission — future work

L5 Orchestration — giu nguyen tu SWE-Agent

Ky thuat: Strategy Pattern + Dependency Injection + Factory Pattern. Moi layer duoc swap qua YAML `config`. Fork tu SWE-Agent (MIT, 15k LOC).

7. PHUONG PHAP — 4 GIAI DOAN THUC NGHIEM

GD1
Taxonomy
5 experts validate, Fleiss kappa ≥ 0.6
Tuan 1–6

GD1.5
Pilot v2
5 cond x 20 tasks x 2 runs = **200 evals**. ~$80.
Tuan 7–8

GD2
Modular Fork
SWE-Agent refactor, verify +/- 3% tren 50 tasks
Tuan 9–11

GD3
Full Ablation
27 cond x 150 tasks = **7,050 evals**
Tuan 12–18

Tai sao pilot? Validate metrics co phan biet duoc giua conditions khong, uoc tinh cost chinh xac, debug pipeline, tinh preliminary effect sizes cho power analysis. **Chong cam ket mu** vao full experiment dat tien.

8. ABLATION DESIGN: 27 CONDITIONS = 3 X 3 X 3

27	150	7,050	~$2.5k
CONDITIONS	TASKS	EVALUATIONS	BUDGET

Factor	Level 1	Level 2	Level 3
Tool Config	Full (12)	Medium (8)	Minimal (5)
Context Strategy	Full History	Sliding Window 50K	Summary/ACC 2K
LLM Backend	Claude Sonnet 4	GPT-4o	DeepSeek-V3

Dataset: 150 tasks SWE-bench Verified, stratified random sampling theo do kho. **So runs:** 10 critical conditions x 3 runs + 17 conditions x 1 run. **Power analysis:** alpha=0.05, power ≥ 0.80.

9. ANOVA 2 CHIEU — PHAN TACH HARNESS VS LLM

"Doc lap voi LLM" **khong phai** nghia la harness va LLM khong tuong tac. Nghia la **co the phan tach** phan variance do harness vs do LLM.

Mo hinh thong ke:

```
Total Variance = Var(Harness) + Var(LLM) + Var(Harness x LLM) + Error

Factor A: Harness config   (9 levels: 3 tool x 3 context)
Factor B: LLM backend      (3 levels: Claude, GPT, DeepSeek)
DV:        Resolve rate

Neu Var(Harness) co y nghia    --> harness CO anh huong doc lap
Neu Var(H x LLM) co y nghia    --> co INTERACTION
eta-squared                    --> % variance cua moi factor
```

Dong gop ly thuyet chinh: Formalize cach **phan tach harness quality khoi LLM quality**. Post-hoc: Tukey HSD pairwise, Cohen's d effect sizes, Bonferroni correction cho 7 metrics.

10. THREATS TO VALIDITY & RISK ANALYSIS

THREATS

Loai	Mitigation
Internal: confounding khi tat tool	Giam 12→8→5, ghi nhan task can tool nao
Internal: LLM non-determinism	Temperature=0, 3 runs cho critical conds
External: Chi Python	Caution khi generalize
Construct: M1.1 phu thuoc annotator	2 annotators, Fleiss kappa

TOP 3 RISKS

Risk	P	Mitigation
API cost vuot budget	Cao	Pilot uoc tinh, DeepSeek re
Docker env fail	TB	Pre-filter, retry x2
Metrics khong phan biet	TB	Pilot phat hien som
DeepSeek API unstable	Thap	Backup Qwen2.5-72B

v2 V1 khong co Threats va Risk Analysis — hoi dong yeu cau bo sung.

11. TIMELINE 24 TUAN & CHI PHI

Giai doan	Tuan	Noi dung	San pham
GD1	1–6	Taxonomy + expert validation	Taxonomy v1.0, kappa report
GD1.5	7–8	**Pilot study** (200 evals)	Metrics validation, cost estimate
GD2	9–11	Fork SWE-Agent + modular refactor	Modular harness v1.0
GD3	12–18	Full ablation (7,050 evals)	Raw data, ANOVA results
GD4	19–22	Analysis + guidelines + viet luan van	Luan van ban thao
GD5	23–24	Review, chinh sua, release toolkit	Final + GitHub release

Chi phi du kien

Pilot: ~$80
Full experiment: ~$2,500
Contingency: ~$500
Tong: $2,500 – 3,100

So sanh v1 / v2

v1: $800–1,200 (thieu thuc te)
v2: $2,500–3,100 (co retry + pilot)
Evaluations: 16,000 → 7,050 (-56%)

12. TRANG THAI CODE — DA TRIEN KHAI

167	9	27	360
TESTS PASS	PHASES DONE	CONDITIONS GENERATED	REAL TRAJECTORIES

TOOLKIT harness_eval/

- `configs/` — 27 conditions 3x3x3
- `metrics/` — 7 metrics M1.1–M3.2
- `parsers/trajectory.py` — SWE-Agent log parser
- `pipeline/analysis.py` — ANOVA, Cohen d, GLMM
- `pipeline/runner.py` — Dry-run + SWE-Agent + Ollama
- `harness/` — 3 ABC + 9 providers + factory
- `cli.py` — info / pilot / run / analyze / convert

DASHBOARD STREAMLIT

- 5 tabs: Config / Run Monitor / Logs / Compare / ANOVA
- Run modes: **ollama** / **dry-run** / **real**
- Parallel ThreadPoolExecutor (4 workers)
- 150 SWE-bench tasks embedded (difficulty-aware)
- Ollama: qwen2.5:7b, deepseek-r1:1.5b
- One-click setup: `start_harness_eval.sh`

13. KET QUA SO BO — ANOVA THUC NGHIEM

Chay tren **trajectories/** hien tai: 360 obs, 18 conditions (2 backends: claude + sonnet)

Source	eta^2	F	p-value	Sig?
Tool	0.0539	10.264	**< 0.001**	***
Context	0.0248	4.723	**0.0095**	**
Backend	0.0025	0.959	0.3282	—
Tool x Context	0.0058	0.550	0.6988	—
Tool x Backend	0.0069	1.314	0.2701	—
Context x Backend	0.0069	1.314	0.2701	—
Error	0.8979	—	—	—

Quan sat so bo:

- H1 huong den **Tool** co effect lon nhat (eta^2 = 0.054 > Context = 0.025)
- H2 huong den **Context** co effect trung binh va **co y nghia thong ke**
- Can them data Backend chua du 3 levels (haiku trong); H4 can full design

Luu y: Data hien tai la dry-run realistic, chua phai SWE-bench that. Ket qua thuc nghiem se co sau GD3.

14. DONG GOP & PUBLISHABILITY

Ly thuyet
Formalize cach phan tach **harness quality khoi LLM quality** qua ANOVA 2 chieu.

Thuc nghiem
Dinh luong dong gop tung component harness lan dau (Tool / Context / Backend).

Toolkit
Framework danh gia **tai su dung** — open-source GitHub, docs, CI/CD.

Y NGHIA THUC TIEN

- **Developer:** biet nen uu tien cai thien component nao cua harness
- **Researcher:** framework chuan de so sanh harness moi
- **Industry:** guidelines thiet ke harness dua tren du lieu

PUBLISHABILITY

Venue	Loai	Phu hop
ICSE 2027 NIER	Conference (New Ideas)	Rat phu hop
MSR 2027	Conference	Phu hop
LLM Agents Workshop (NeurIPS/ICML)	Workshop	Phu hop

Q&A

Xin cam on quy thay co va hoi dong
da lang nghe va phan bien

Resources:

`NCKK-Docs/de-tai/de-cuong/DE-CUONG-HarnessEval-v2.md`
`NCKK-Docs/de-tai/harness-eval/` — toolkit + 167 tests
`plans/Plan-Coding-HarnessEval.md` — session log